

One Interval, Two Threshold Claims

Unit 3 · Topic 3.4 · TODAY'S PROBLEM

Suppose a statewide activities association has 10,000 member students and is considering requiring reusable food containers at its tournaments. In a simple random sample of 300 member students, 204 favor the requirement. A sponsor claims that at least 65% of all member students favor it; an event organizer makes the more modest claim that more than 60% favor it.

Kai: “The sample has $204/300 = 68\%$ in favor, above both thresholds, so it supports both claims.”

Mira: “The point estimate alone cannot settle both claims. We should construct a confidence interval and compare all of its plausible values with each threshold.”

Sample counts

Group	N	n	Favor	Other
Members	10,000	300	204	96

FIRST TAKE (5 min, on your sheet)

Do the data appear to support both claims, only one, or neither? Cite one number and commit before calculating the interval.

- DOK 1** (a) Define the population proportion p and calculate the sample proportion $\hat{p}$.
- DOK 2** (b) Verify the conditions, construct a 95 percent confidence interval for p , and interpret the interval in the association's context.
- DOK 3** (c) In 3–4 sentences, decide which threshold claim the interval supports and whose reasoning is better. Compare the lower endpoint with 0.60 and 0.65. Explain why a plausible 0.65 is different from establishing at least 0.65.

Optional review: [follow-along](#) (scan the code)

Work (a)–(c) from the rules on your sheet. Turn in your sheet.

